Combination circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Reduce a circuit by identifying genuine series and parallel groups.

SUCCESS CRITERIA - I can...

1. Identify nodes before grouping resistors.
2. Reduce one group at a time.
3. Check current and voltage conservation in the original circuit.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “parallel circuit” to explain your model or result. Sentence starters are optional.

series: connected along one unbranched current path

parallel circuit: branches connected across the same two nodes

node: a shared electrical connection point

01 Warm-up: bring back an earlier idea

Measured (I, V) pairs are (0.10, 3), (0.20, 6), (0.30, 9) in A, V.
Find resistance and explain the evidence for an ohmic model.

Word help: resistance = voltage divided by current for a stated operating condition

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $R_{series} = R_1 + R_2$; $1/R_{parallel} = 1/R_1 + 1/R_2$; $V = IR$. Reduce one known group at a time. Check nodes before calling components series or parallel.

Worked reasoning: A 2 ohm resistor is in series with two 4 ohm resistors in parallel. The parallel pair equals 2 ohms; total 4 ohms. A 12 V supply gives 3 A total.

Your turn: Draw the network and label the two series sections before splitting the branch current.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

Combination circuits

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A 2 Ω resistor is in series with two parallel 6 Ω resistors. **Draw** the arrangement and circle the parallel group.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A 10 V supply powers a 2 Ω resistor in series with two parallel 6 Ω resistors.
Label the network.
Find parallel equivalent resistance, total current, series voltage drop, branch voltage and each branch current.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student adds all three resistor values because they are drawn side by side. What should determine the grouping instead?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two targets each have range 600 m but different radial velocities.
Explain how they can have equal echo delays yet different Doppler shifts.
Label the outgoing and returning paths.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

Combination circuits

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Replace the two 6 Ω branches with 3 Ω and 6 Ω. Keep the 2 Ω series resistor and 10 V supply.
Find total current.
Label the circuit and show the current path or paths.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 3 ohm series resistor feeds parallel 6 and 3 ohm branches from 15 V.
Find total current, voltage across parallel section and each branch current.
Label the circuit and show the current path or paths.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3